

Notes on Radio Evolution of Galaxies

Anurag Tamuli

PRL '26

Abstract

The primary source for these notes has been *Extragalactic Astronomy and Cosmology* by *Peter Schneider*. These notes were made during my summer internship at PRL Thaltej from May 11 to July 10, 2026.

Contents

1	Useful Terminology and Concepts	2
1.1	Basic Terminology	2
1.2	The Magnitude Scale	2
1.2.1	Apparent Magnitude	2
1.2.2	Filters and Colour	3
2	The World of Galaxies	4
2.1	Classification of Galaxies	5
2.1.1	Morphological Classification	5

Chapter 1

Useful Terminology and Concepts

1.1 Basic Terminology

Luminosity (L): The total amount of electromagnetic **energy emitted per unit time** by an astronomical object. L_B and needs to be mentioned when we bring up colour filters.

Radiant Flux Density (S): The total amount of electromagnetic **energy per unit area per unit time** by an astronomical object. In other words, it is the luminosity per unit area. According to this definition, at a distance r from a star (say):

$$S = \frac{L}{4\pi r^2}$$

1.2 The Magnitude Scale

Our eyes judge brightness according in a logarithmic fashion, similar to hearing. This means that what we perceive to be twice as bright is actually emitting a lot more than just twice the radiant energy. Due to this, optical astronomy uses ‘magnitude’ as a scale of brightness.

1.2.1 Apparent Magnitude

For two sources with fluxes¹ S_1 and S_2 , their *apparent magnitudes* m_1 and m_2 are defined as:

$$m_1 - m_2 = -2.5 \log\left(\frac{S_1}{S_2}\right)$$

¹Here, we are talking about radiant flux density, defined by energy emitted per unit area per unit time, i.e. luminosity per unit area.

This means that a brighter object will have a smaller apparent magnitude. The factor of 2.5 is chosen to most accurately match the historically calculated magnitudes.

1.2.2 Filters and Colour

Optical observations are performed with a combination of filters and detectors. Since the spectral energy distribution is not necessarily uniform over all wavelengths, apparent magnitudes are defined for each of these filters, i.e. apparent magnitude is a function of the wavelength of light being considered.

$$m = m(\lambda) \equiv m_X \equiv X$$

where X is the filter for a specific wavelength λ .

Chapter 2

The World of Galaxies

Ever since the Hubble Deep Field image, the whole of humanity has set their sight to discover what lies far beyond they could ever comprehend. When dealing with extragalactic astronomy, the most notable entities are arguably other galaxies, and this is a chapter all about them.



Figure 2.1: The Hubble Deep Field image produced by *Edwin Hubble* changed the world forever. All the dots in the image are galaxies, not stars.

The light from 'normal galaxies' is mainly the sum of all the light emitted by the stars present within it.

2.1 Classification of Galaxies

Galaxies **dafds**

2.1.1 Morphological Classification

On the basis of shape or morphology, galaxies have been historically classified into the following categories:

1. **Elliptical Galaxies:** Their isophotes¹ resemble ellipses. They are further subclassified according to the **ellipticity** of the isophotes: $\epsilon = 1 - b/a$, where a, b are the semi-major and semi-minor axes. Then we define $N = 10\epsilon$ which gives us the representation EN for elliptical galaxies. An E7 galaxy is an elliptical galaxy with $b/a = 0.3$.
2. **Spiral Galaxies:** Their isophotes look like spirals.

¹Isophotes are contours along which the surface brightness is constant.